

Lab 34 — Pandas Grouping and Aggregation

Aim

To group data by a categorical column and compute aggregate statistics for each group.

Algorithm / Procedure

1. Import the Pandas library.
2. Create a DataFrame containing class and marks.
3. Group the data using the class column.
4. Calculate the mean marks for each class.
5. Calculate multiple statistics such as mean, maximum, and minimum.
6. Display the results.

Program

```
import pandas as pd
df=pd.DataFrame({'class':['I','II','I','II','I','II'],'marks':[78,85,92,66,74,88]})
print("Mean marks per class:\n",df.groupby('class')['marks'].mean())
print("Multiple aggregates:\n",df.groupby('class')['marks'].agg(['mean','max','min']))
```

Output

Mean marks per class:

class

I 81.333333

II 79.666667

Name: marks, dtype: float64

Multiple aggregates:

mean max min

class

I 81.333333 92 74

II 79.666667 88 66

Result

Pandas grouping and aggregation were successfully performed. Mean, maximum, and minimum marks were calculated for each class.